Why a whiteboard-led AI tutor?

The case for a shared learning surface that adapts to how students think.

01 The thesis

From classrooms to research labs, some of the best learning happens on a shared surface. **Y aims to bring that experience to AI tutoring:** a place where students and AI can work through problems together, rather than exchanging answers in isolated chat turns. The longer-term vision combines multimodal tutoring with a self-improving representation of each learner's knowledge and understanding.

THE PROPOSED LEARNING LOOP

Student input → **Shared surface** → **Learner model** → **Adaptive tutor**

Write, draw, ask — Think together — Track understanding — Tailor next step

Each interaction can refine what the tutor understands and does next.

02 Why a whiteboard?

Active participation

Students can attempt, revise and explain their thinking while the tutor responds to their intermediate steps.

Visual freedom

A shared space supports equations, annotations and diagrams that develop alongside the explanation.

Across disciplines

The same surface can support math, physics, chemistry, biology, economics and more.

> **The interface is a hypothesis:** shared visual context may make multistep, visual problem-solving more effective than chat alone.

WHAT SHARED VISUAL CONTEXT COULD CHANGE

Illustration, not a live feature

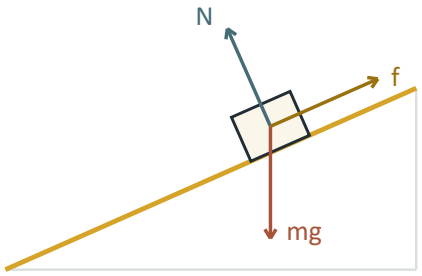
Text exchange

Student Why does friction point up the slope?

Tutor It opposes the block's tendency to slide.

Explanation lives in the message thread.

Shared working surface



The student and tutor can mark the same diagram.

03 How it compares with other interfaces

The goal is not to replace chat or voice. It is to make the right medium available for each task, with a shared visual workspace when the learner needs it.

Chat	Fast, familiar; less natural for spatial reasoning and joint working.
Voice	Conversational and hands-free; harder to convey complex visual relationships.
Video / diagrams	Clear illustrations; typically less interactive than developing them together.
Document-grounded AI	Keeps teaching within course materials; retrieval alone does not reveal misconceptions.
Shared whiteboard	Supports joint working, annotations, explanations and student participation.

04 India: tablets today, mobile tomorrow

Many schools already use tablets for learning and assignments, while others are introducing them. This gives a whiteboard-led tutor a natural entry point for students tackling equations and diagrams, especially in JEE/NEET and undergraduate courses. **For broader reach, the same tutor should also work on phones without a stylus.**

Tablet	Desktop	Phone
Write and draw directly	Use a shared workspace	Chat, voice, photos of work

05 Early exploration and current stage

I briefly explored a retrieval-augmented approach to grounding explanations in external learning material. Uploading notes, slides, textbooks or video references could support syllabus-aware teaching and quizzes. But **having the right content is different from understanding the student.** The core product question is whether interaction evidence can improve future tutoring.

EXPLORED	**Lightweight RAG for external course context** An initial technical exploration, not a formal form-factor experiment.
AVAILABLE NOW	**Early core tutoring experience** The demo supports the initial teaching experience, not the full long-term vision.
IN DEVELOPMENT	**Joint drawing and evolving learner representation** The proposed whiteboard interaction and self-improving learning loop are being integrated.

What remains to be proven: whether a whiteboard-led experience improves understanding, participation and continuity over chat or voice alternatives. I have not yet run controlled comparisons.

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